

Preliminary Research & Methodology Report

Title: Integrated AI Framework for Sustainable Restaurant Operations: Demand Forecasting via Aspect-Based Sentiment Analysis and Resource Optimization

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1. Executive Summary & Problem Statement

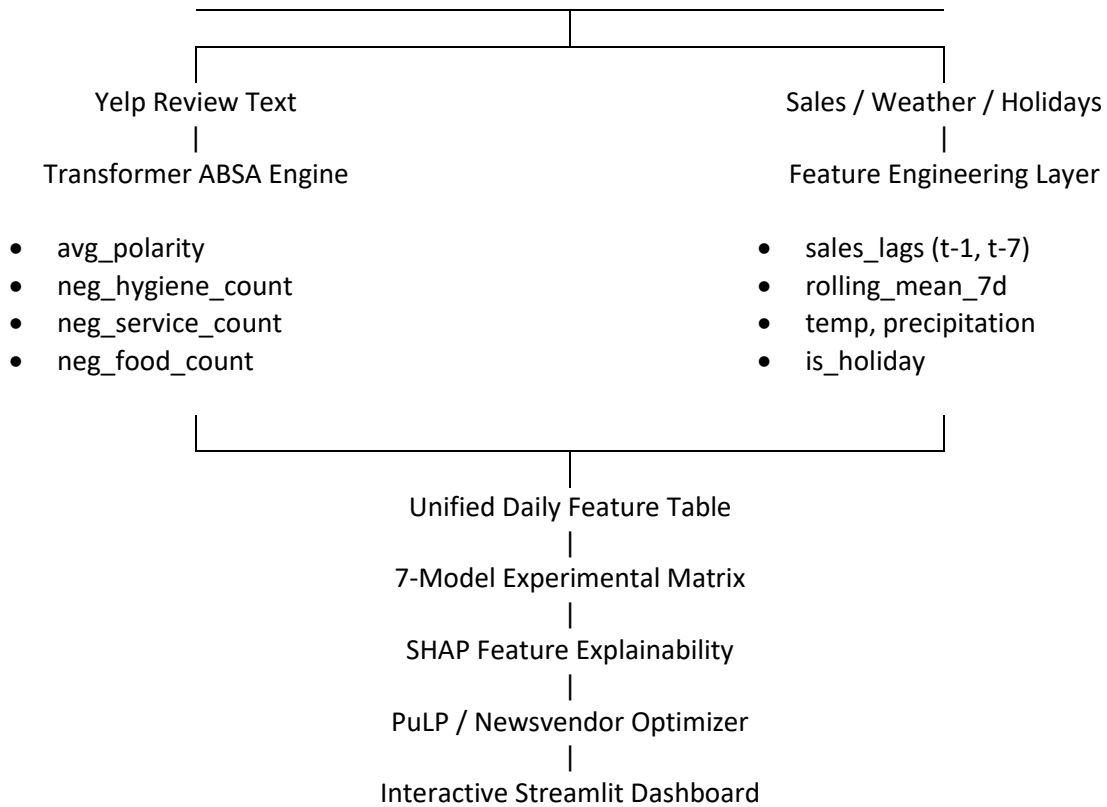
Restaurant managers work under conditions of constant uncertainty about customer flow, stock movement, and staffing needs. The traditional demand forecasting techniques only consider historical data, weather changes, or national holidays. On the contrary, existing customer feedback analysis systems convert textual reviews into sentiment scores but do not incorporate the information into decision making. The overall star rating is not enough for getting an idea of how operations run; a bad rating is unable to show which of the reasons caused a decrease in sales.

The proposed pipeline in this research is aimed at addressing the above issue. The analysis of specific sentiments regarding hygiene, service quality, food, ambience and prices of the restaurant through customer feedback will help us determine whether more accurate next-day demand forecasts can be achieved using the fine-grained sentiment metrics in comparison to non-textual models. Moreover, these forecasts will be used for calculating staff scheduling and stocking requirements.

2. Theoretical Framework & Literature Mapping

The proposed system synthesizes three primary domain areas:

1. **Univariate & Structured Forecasting:** Grounded in classical time-series techniques including ARIMA (Box et al., 1976), Prophet (Taylor & Letham, 2018), Gradient-Boosted Decision Trees via XGBoost (Chen & Guestrin, 2016), and Recurrent Neural Networks via LSTMs (Hochreiter & Schmidhuber, 1997).
2. **Aspect-Based Sentiment Analysis (ABSA):** Utilizing the SemEval-2014 aspect taxonomy (Pontiki et al., 2014) alongside pretrained transformer encoders (BERT/RoBERTa; Devlin et al., 2019; Liu et al., 2019) fine-tuned according to the domain post-training methodology of Xu et al. (2019).
3. **Review Sentiment & Demand Modeling:** Extending foundational findings by Ye et al. (2009), which demonstrated the impact of overall online ratings on customer demand, into fine-grained aspect granularity.



3. Data Integration & Methodological Setup

3.1 Datasets & Data Fusion Layer

- **Customer Reviews:** Extracted from the public Yelp Open Dataset filtered for restaurant entities, supplying raw review text, timestamps, and establishment metadata.
- **Demand / Sales Series:** Daily sales data sourced from public Kaggle store item demand repositories. Where explicit joint sales and review records for identical entities are unavailable, a controlled, transparent synthetic linkage will be generated matching realistic review sentiment distributions to sales series.
- **External Features:** Historical daily weather metrics (temperature, precipitation) acquired via the Open-Meteo API, paired with national holiday indicators via the Python `holidays` package.

3.2 The 7 Experimental Conditions

To explicitly isolate the marginal predictive value of fine-grained aspect sentiment, seven models will be evaluated on an identical 80/20 time-ordered train/test split:

<i>Model ID</i>	<i>Architecture</i>	<i>Input Feature Composition</i>	<i>Experimental Purpose</i>
<i>Model 1</i>	ARIMA	Sales lags (t-1,t-7)	Univariate baseline floor
<i>Model 2</i>	Prophet	Trend, seasonality, public holiday flags	Industry-standard seasonal baseline
<i>Model 3</i>	XGBoost	Lags, rolling stats, weather, holiday indicators	Non-text structured data baseline
<i>Model 4</i>	LSTM	7-day historical sales sequence	Deep sequence model baseline
<i>Model 5</i>	XGBoost	Model 3 features + avg_polarity	Evaluates coarse overall sentiment
<i>Model 6</i>	XGBoost	Model 3 features + negative aspect counts	Evaluates fine-grained aspect counts
<i>Model 7</i>	XGBoost	Model 3 features + avg_polarity + aspect counts	Proposed combined approach

4. Expected Outcomes & Evaluation Plan

4.1 Forecast Accuracy & SHAP Feature Attribution

- **Primary Metrics:** Models will be benchmarked using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) on held-out test periods.
- **Hypothesized Result:** Model 7 (Full ABSA) is expected to outperform non-text baselines (Model 3) and coarse sentiment models (Model 5) by capturing critical operational warning signals (e.g., negative service or hygiene spikes).
- **Explainability:** Model 7 predictions will be passed to `shap.TreeExplainer` to extract SHAP feature importance rankings, quantitatively verifying which specific review aspect drives demand shifts.

4.2 Downstream Decision Support (Optimization Layer)

- **Staff Scheduling:** Formulated via `PuLP` linear programming to minimize shift costs while satisfying daily predicted customer demand constraints.
- **Inventory Allocation:** Computed via a classical Newsvendor optimization policy to balance stockout costs against waste overage costs.
- **Dashboard Output:** A prototype Streamlit interface will display daily demand forecasts, SHAP contribution breakdown plots, optimal shift/inventory purchase orders, and "what-if" scenario sliders.

5. Execution Timeline & Next Steps

1. **Phase 1 (Weeks 1–2):** Finalize Kaggle dataset downloads, execute RoBERTa fine-tuning for daily aspect extraction, and build the merged data table.
2. **Phase 2 (Weeks 3–4):** Execute the 7-model training loop in Google Colab (T4 GPU runtime), log test metrics (RMSE/MAE), and generate SHAP summary plots.

3. **Phase 3 (Weeks 5–6):** Develop the `PuLP` staff scheduling optimization scripts and build the interactive Streamlit user dashboard.
4. **Phase 4 (Weeks 7–8):** Compile final execution logs into LaTeX figures, write up empirical findings, and submit the complete paper.